

Occupancy modelling with bioacoustics

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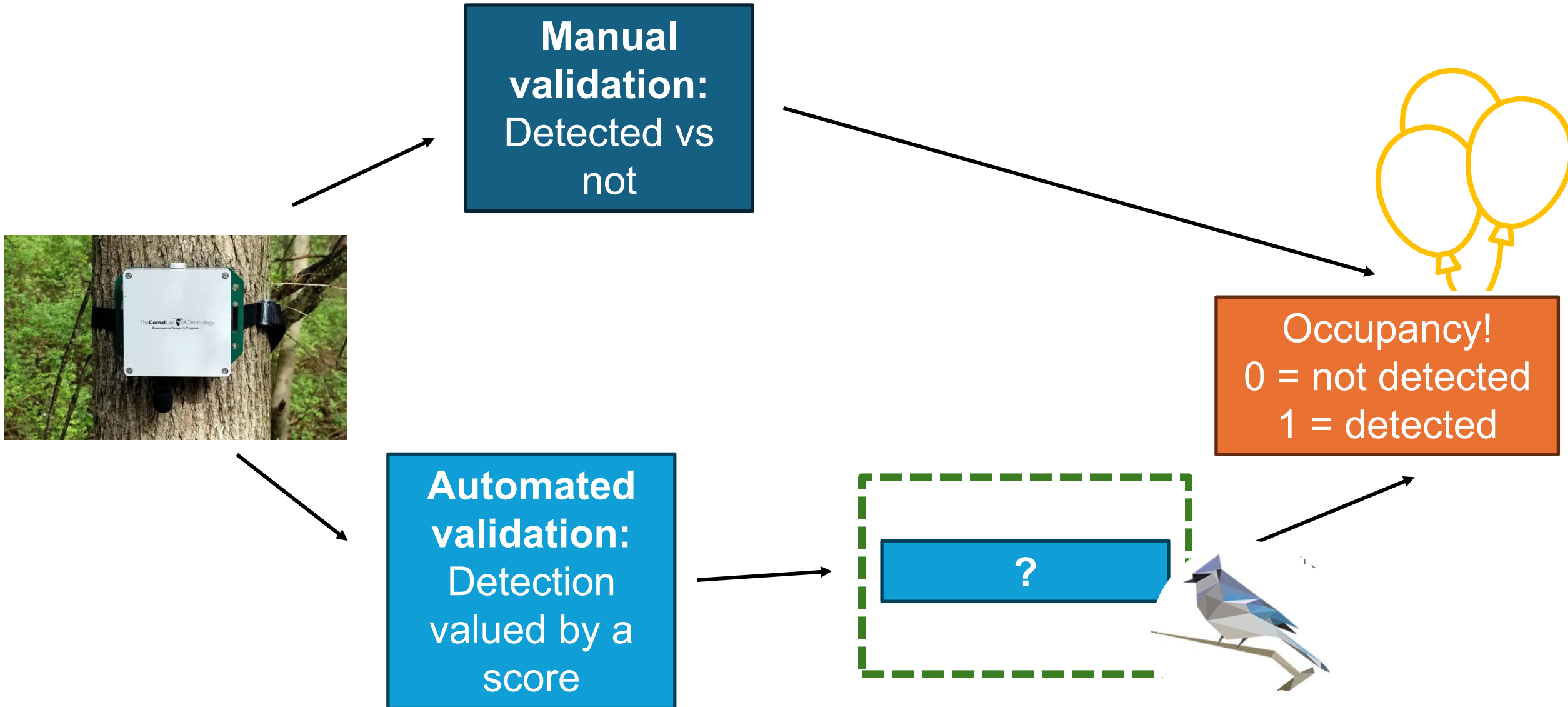
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Overview

- Creating a detection history from BirdNET output
- Why do we want to use occupancy models?
- How do occupancy models work?
- Study design for occupancy modelling
- Relaxing assumptions (multi-species, multi-season, FP)
- All of this is within the context of bioacoustics!

Making acoustic data ready for occupancy



BirdNET Thresholding for detection/non-detection data

- Manually validate a subset of your BirdNET continuous output

File name	confidence score	logit score
987_G080_V1_C6174_U2_20210622_060002Z.flac	0.987	4.330
993_G090_V1_C6728_U1_20210626_190000Z.flac	0.993	4.955
103_G064_V1_C5306_U2_20210603_010000Z.flac	0.103	-2.164
114_G076_V1_C5380_U1_20210613_070003Z.flac	0.114	-2.051

BirdNET Thresholding for detection/non-detection data

- Fit logistic regression models to the data
 - **Predictor** - BirdNET output (confidence and logit scores)
 - **Response** - prediction outcome (0 = incorrect, 1 = correct)

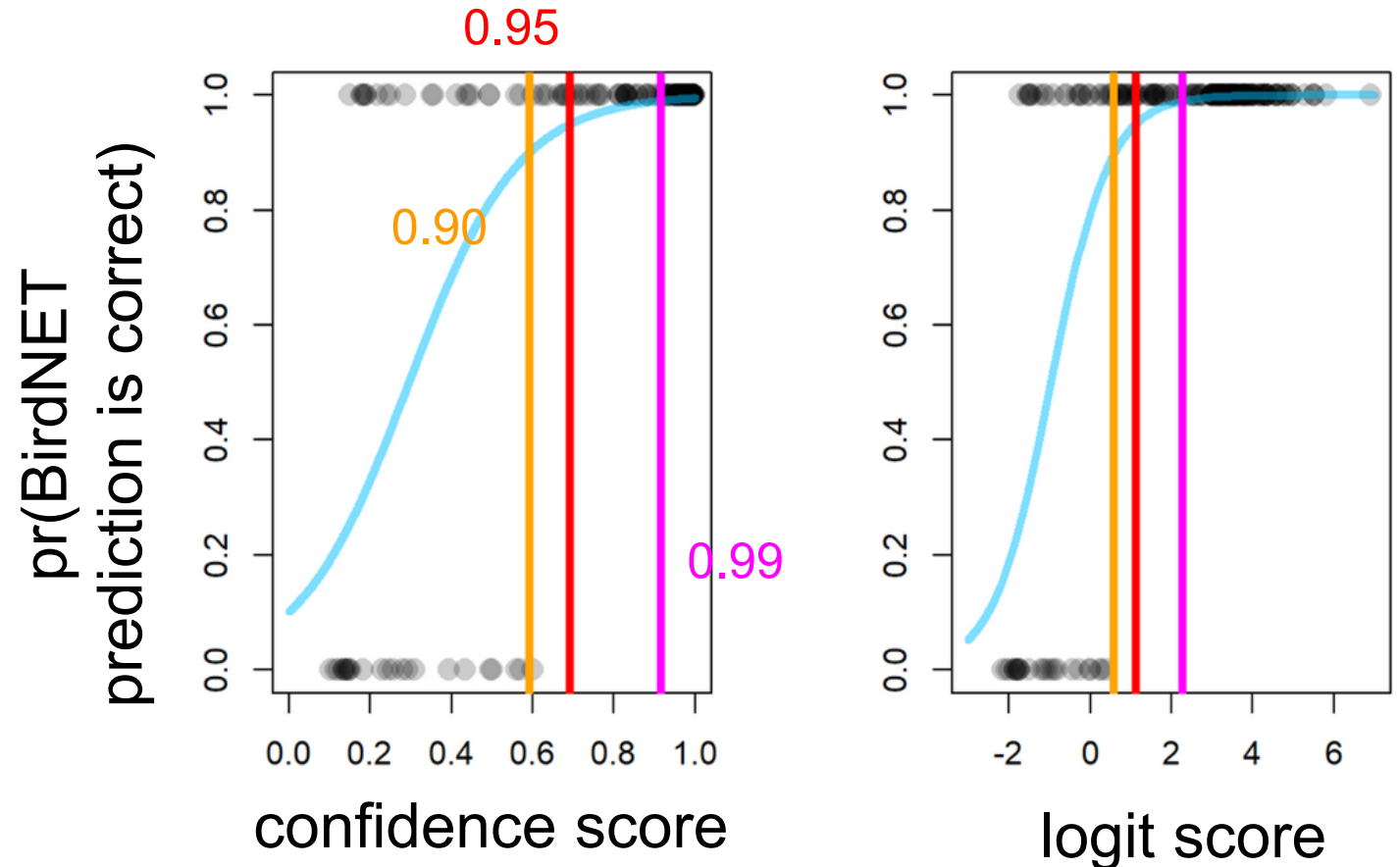
$$\text{logit}(\text{correct}) = \beta_0 + \beta_1 \text{score}$$

correct
1
1
0
0

confidence score
0.987
0.993
0.103
0.114

BirdNET Thresholding for detection/non-detection data

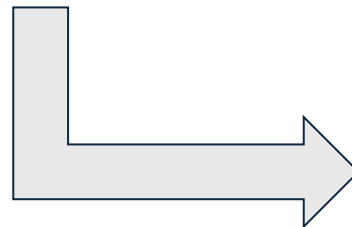
- Solve the logistic equations for the desired probability thresholds
- e.g., 90%, 95%, 99% probability a detection is a true positive



BirdNET Thresholding for detection/non-detection data

- Choose a threshold, binarize the data, create a detection history
 - Considerations for a threshold (precision vs recall, sites lost)

File name					logit score
987	G080	V1_C6174_U2	20210622	060002Z.flac	4.330
993	G090	V1_C6728_U1	20210626	190000Z.flac	4.955
103	G064	V1_C5306_U2	20210603	010000Z.flac	-2.164
114	G076	V1_C5380_U1	20210613	070003Z.flac	-2.051



Site	visit1	visit2	visit3
G080	1	0	1
G090	1	1	1
G064	0	1	0
G076	0	0	0

CODING BREAK

Let's walk through how to use BirdNET thresholding for creating detection histories in R

Goals:

- Learn one method to translate BirdNET output to a detection history
- Understand how to consider precision/recall when selecting a BirdNET threshold

Marc Kéry
J. Andrew Royle



APPLIED HIERARCHICAL MODELING IN ECOLOGY

Analysis of distribution, abundance and
species richness in R and BUGS

Volume 1
Prelude and Static Models



Marc Kéry
J. Andrew Royle



APPLIED HIERARCHICAL MODELING IN ECOLOGY

Analysis of distribution, abundance and
species richness in R and BUGS

Volume 2
Dynamic and Advanced Models



SECOND EDITION

OCCUPANCY ESTIMATION AND MODELING

INFERRING PATTERNS AND DYNAMICS OF SPECIES OCCURRENCE

Darryl I. MacKenzie, James D. Nichols, J. Andrew Royle,
Kenneth H. Pollock, Larissa L. Bailey, James E. Hines

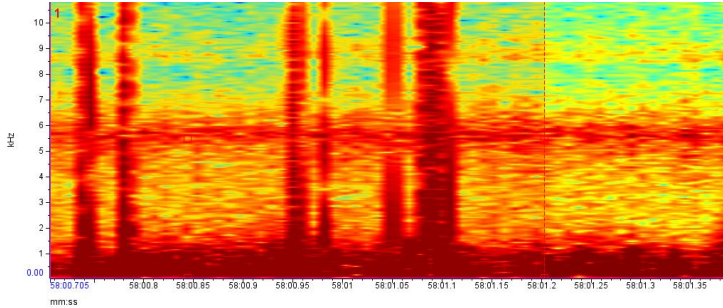


Surveying wildlife populations

- Individual identification is hard! Presence is *relatively* easy
- Surveys yield detection/non-detection data
 - **Non-detection $\neq$ absence! ...why?**
- Animals are difficult to detect and humans are imperfect
- Hard to understand a population with low-quality info



Imperfect detection is a reality...not an anomaly



Credit: Marthazillig.github.io

Which population is bigger?

Site 1

Site 2

Site 3

Site 4

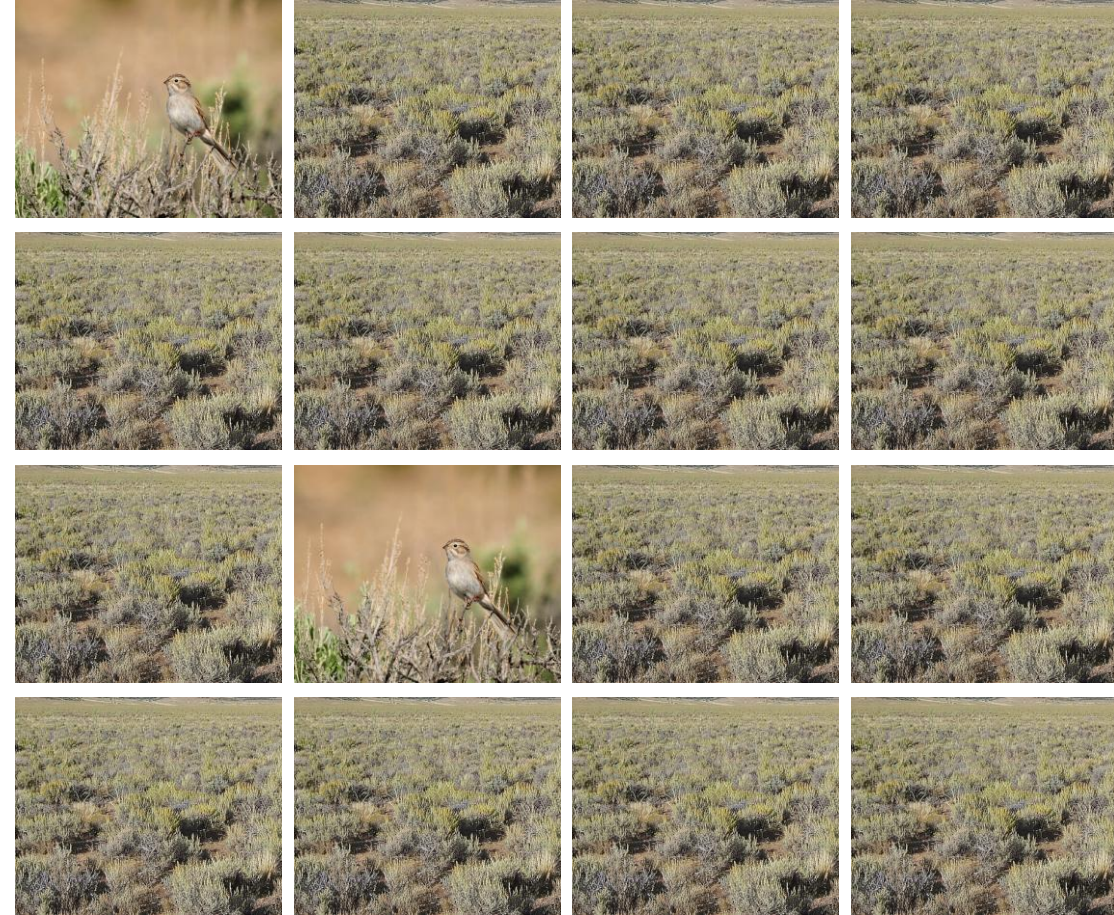


Site 1

Site 2

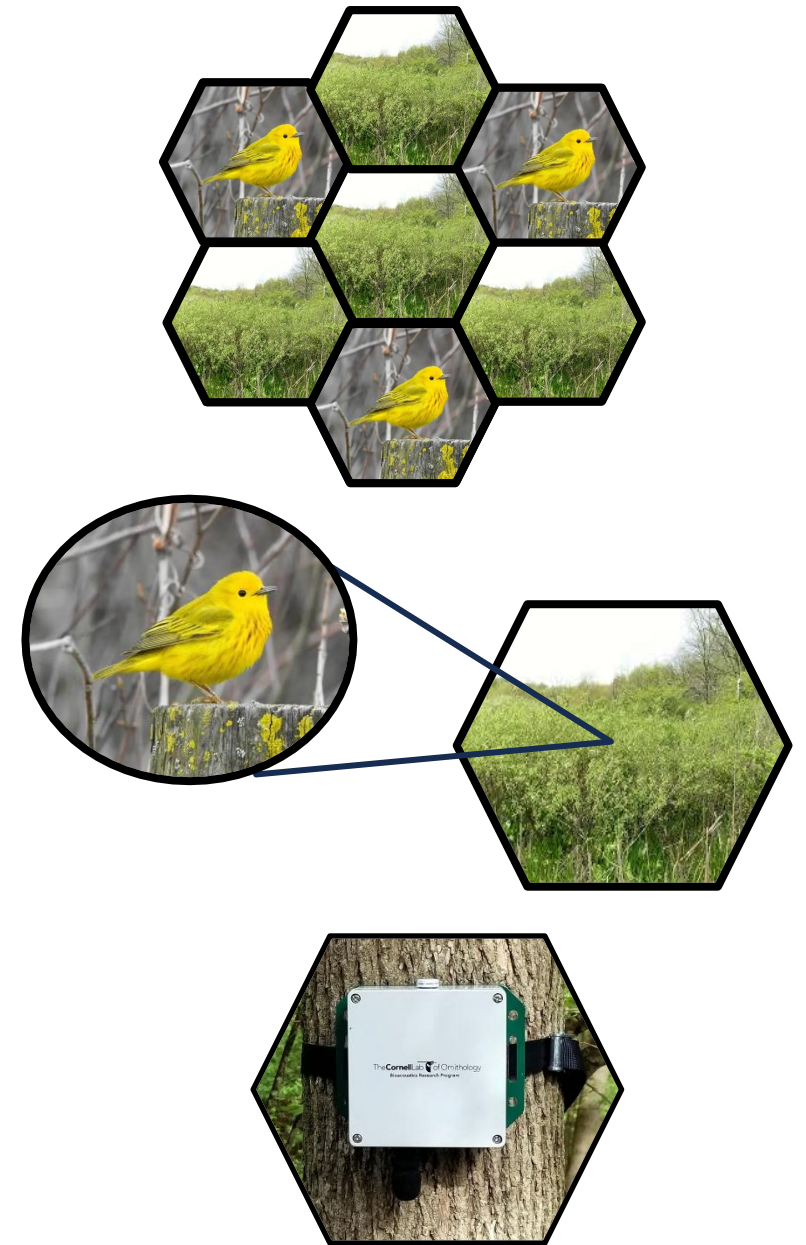
Site 3

Site 4



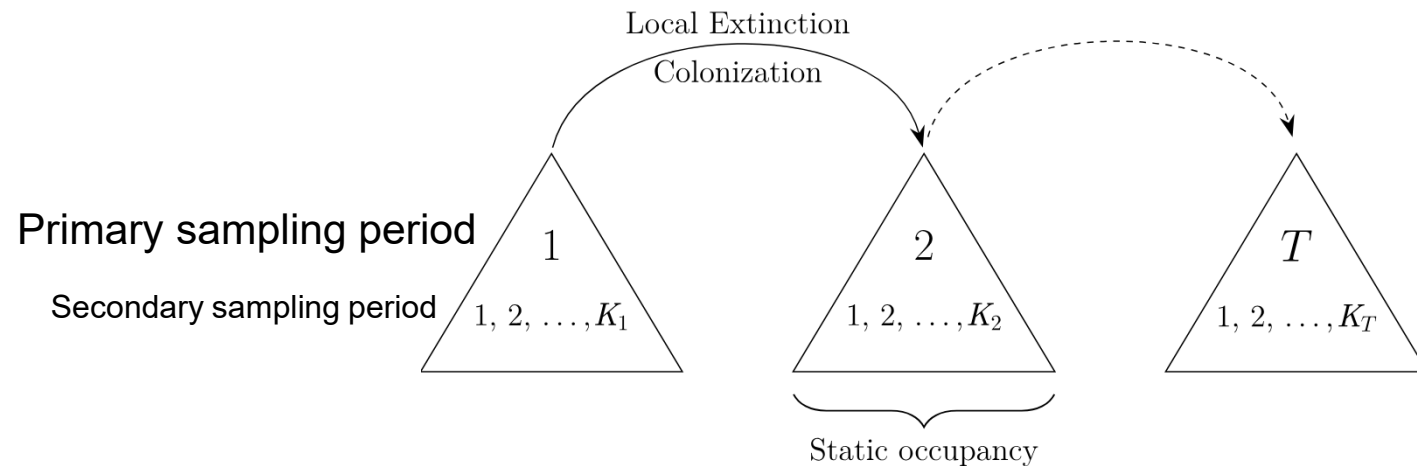
A few quick definitions:

- Naïve occupancy – The proportion of sites the species occurs across all sampled sites
- Occupancy (Ψ) – The probability that a species is present within a sampling unit during a primary sampling event
- Detection (p) – The probability that we observe a species if it occupies a sampling unit during a primary sampling event
- Sampling unit – A discrete unit of space that we can meaningfully survey



A few quick definitions:

- Primary sampling period – A time period where we assume that the occupancy status of a given species does not change
- Secondary sampling period – A time period that defines the repeat time visits where we have the chance to observe the species



What is an occupancy model?

- **Occupancy model:** mathematical representation of the probability that an area is occupied by a focal species
- (At least) two parameters:
 - Occupancy: probability that a location is occupied by a species
 - Property of a *site*
 - Detection: probability of observing a species, given that it is present
 - Property of an *individual survey* (one of your repeated visits)

Occupancy:
Animal at your
survey sites?

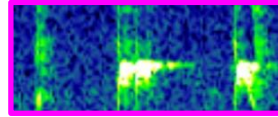
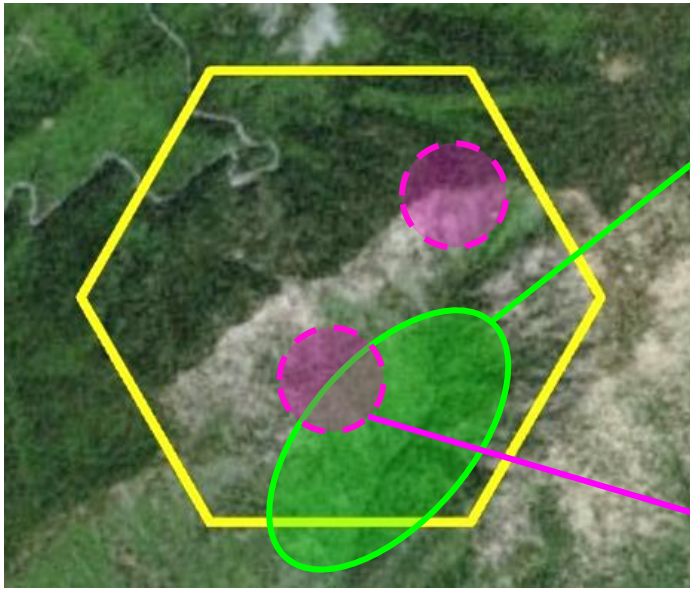
1
0
1
1

Detection:
animal observed at a
given site?

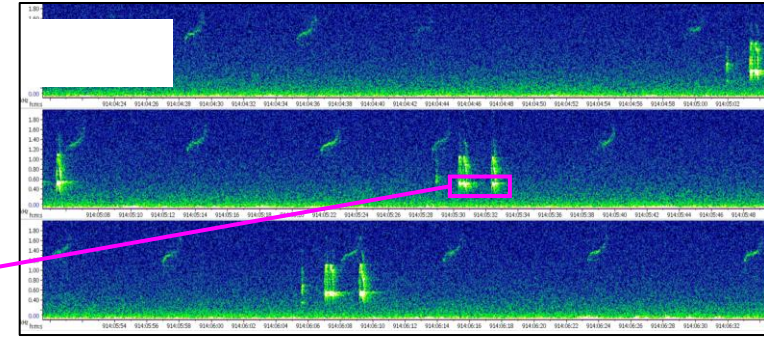
Tyre et al. 2001 Inferring process from pattern: can territory occupancy provide information about life history parameters? *Ecol. Apps.*
MacKenzie et al. 2002. Estimating site occupancy rates when detection probabilities are less than one. *Ecology*

What are “**occupancy**” and “**detection**”?

State Process: Ecological



Sampling Process: Methodological



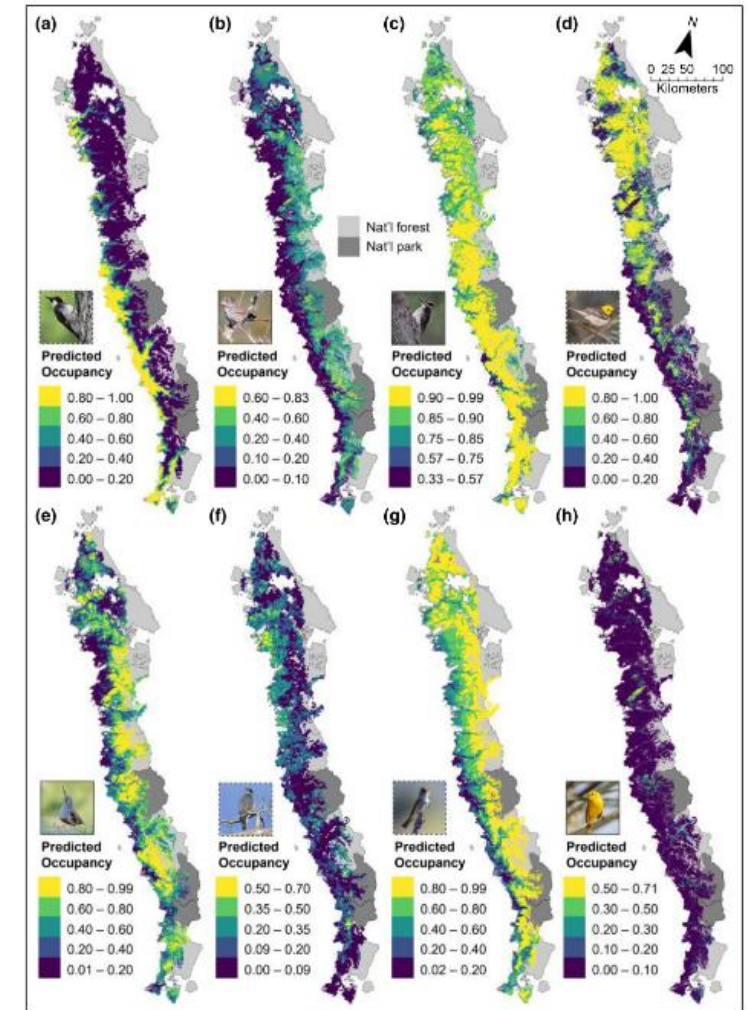
Detection Histories

1	0	1
0	0	1
0	0	0
0	0	0
1	1	1
1	1	0

} **occupancy**
and **detection**

What can we learn from estimating occupancy?

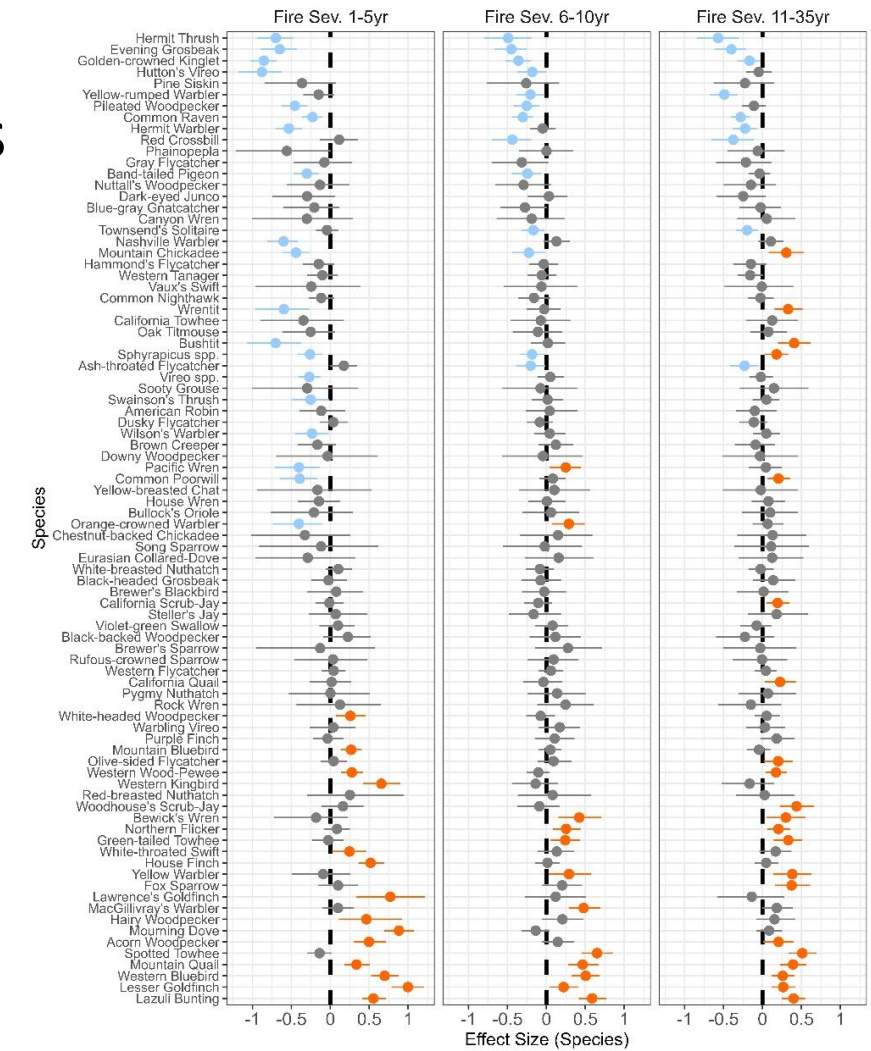
- Species distributions and geographic ranges



Brunk et al., 2025

What can we learn from estimating occupancy?

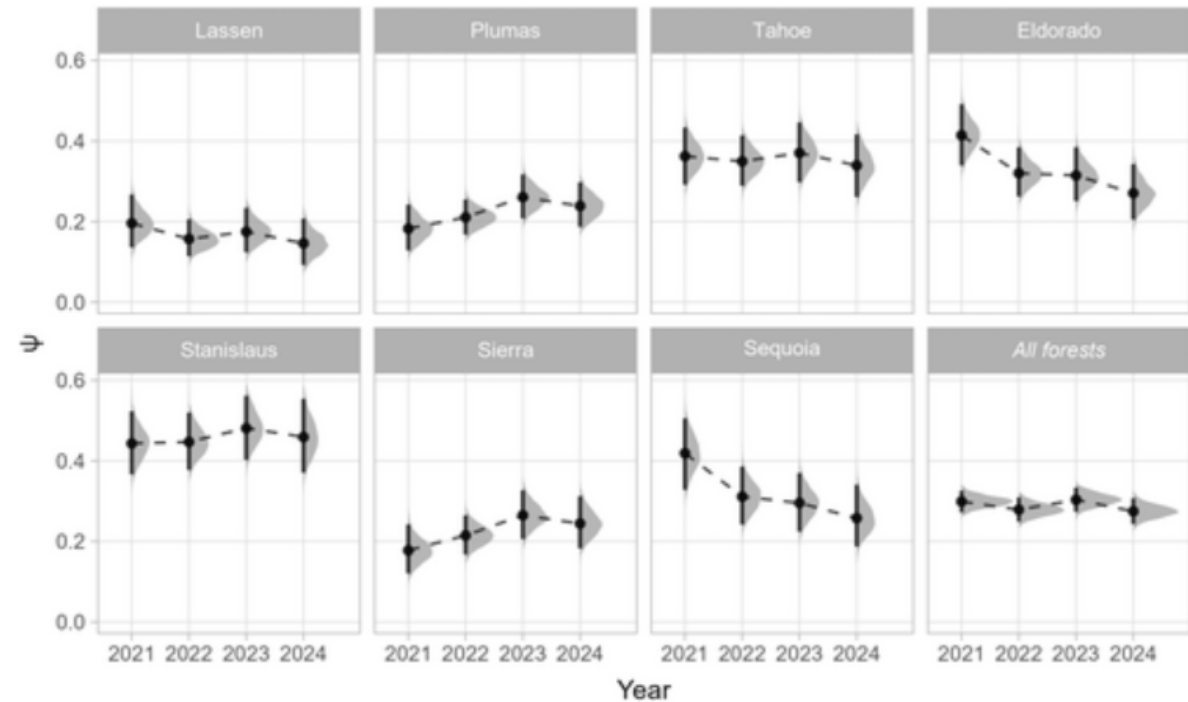
- Species distributions and geographic ranges
- **Species habitat use and relationships**



Keyser et al., In-review

What can we learn from estimating occupancy?

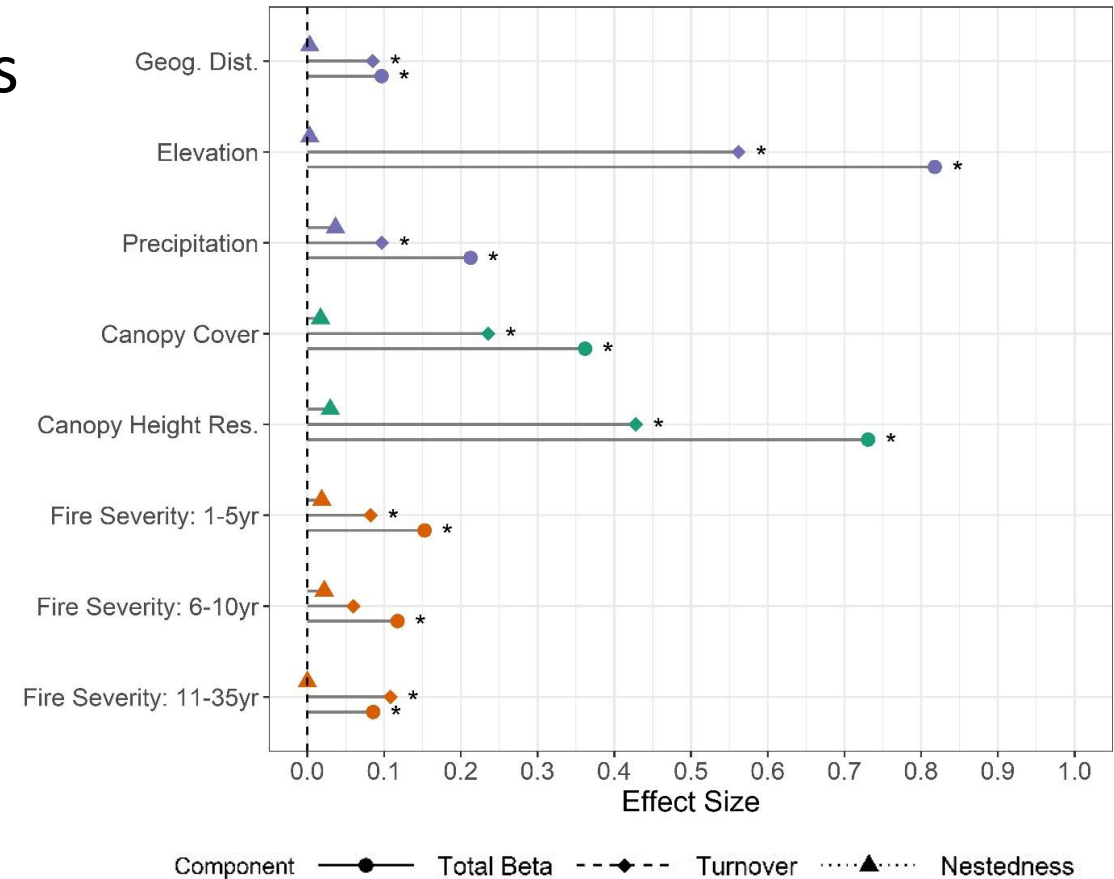
- Species distributions and geographic ranges
- Species habitat use and relationships
- **Large-scale monitoring**



Winiarski et al. 2026

What can we learn from estimating occupancy?

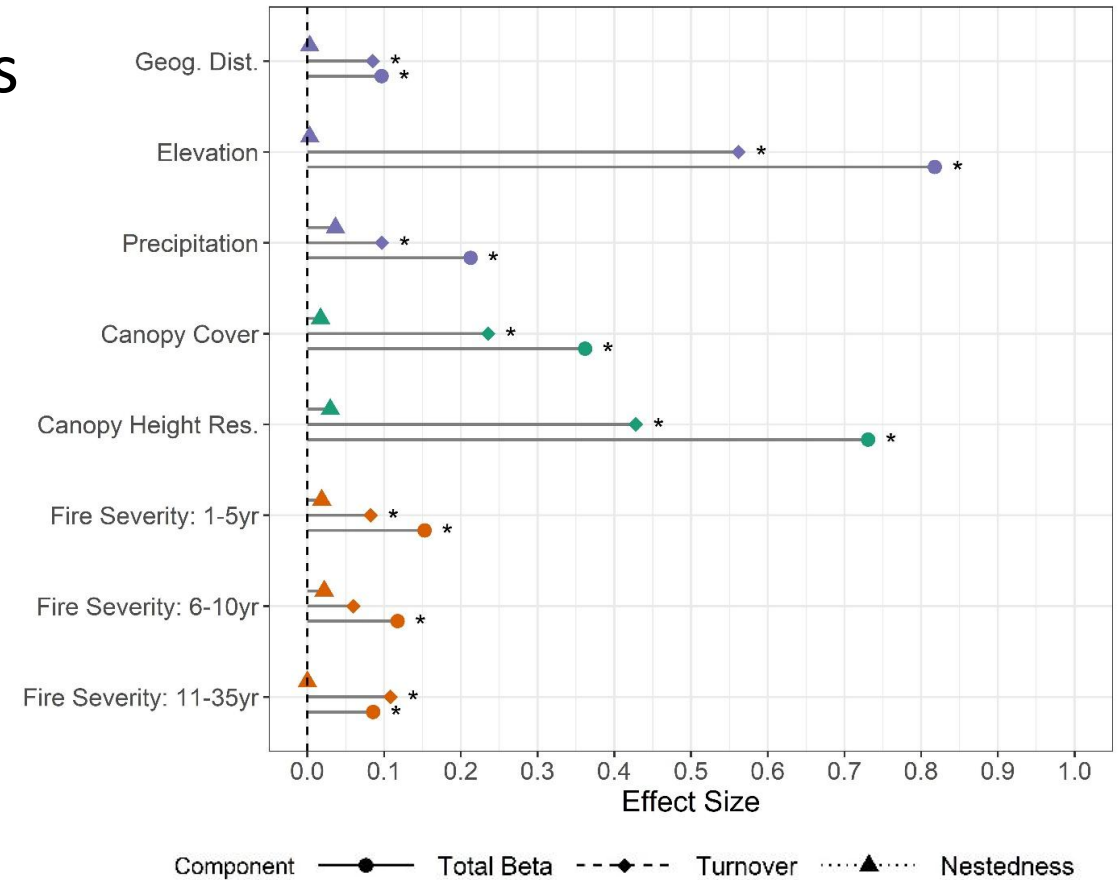
- Species distributions and geographic ranges
- Species habitat use and relationships
- Large-scale monitoring
- **Community analysis**



Keyser et al., In-review

What can we learn from estimating occupancy?

- Species distributions and geographic ranges
- Species habitat use and relationships
- Large-scale monitoring
- Community analysis
- **Metapopulation dynamics**
- **Disease modeling**
- **Human-wildlife conflict zones**
- **Species interactions**



Keyser et al., In-review

Simplest form of site-occupancy model

Ecological (State) Process

$$z_i \sim \text{Bernoulli}(\psi)$$

Observation Process

$$(y_i | z_i = 1) \sim \text{Bernoulli}(p)$$

$$(y_i | z_i = 0) \sim \text{Bernoulli}(q)$$

$$y_{ij} \sim \text{Bernoulli}(z_i p + (1 - z_i) q)$$

$$y_{ij} | z_i \sim \text{Bernoulli}(z_i p)$$

Simplest form of site-occupancy model

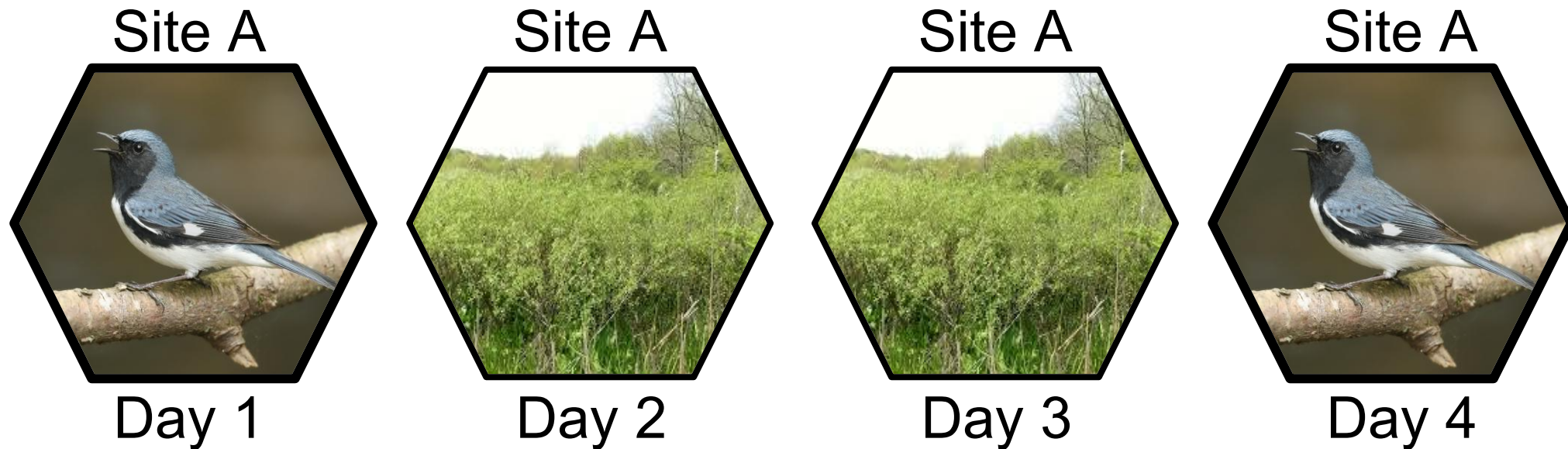
State Process Model: $z_i \sim \text{Bernoulli}(\psi)$ where z_i is the *latent* state of species presence absence at site i and ψ is occupancy probability

In other words...the true presence or absence of site i is a Bernoulli random trial with probability ψ

Observation Model: $y_{ij}|z_i \sim \text{Bernoulli}(z_i p)$ where y_{ij} is the observed presence/absence record at site i and survey j , z_i is the true, latent state of the sites, and p is the detection probability

Or...the observed state given the true state is a Bernoulli random trial with probability equal to detection probability when the site is occupied

What is a capture history?



$$H = [1, 0, 0, 1]$$

Observed data likelihood

Assume a detection history of [1,0,0,1]

What is $\Pr(H = 1001)$?

$$\Pr(H = 1001) = \psi p_1 (1 - p_2) (1 - p_3) p_4$$

What about $\Pr(H = 0000)$?

$$\Pr(H = 0000) = \psi \prod_{j=1}^4 (1 - p_j) + (1 - \psi)$$

$$= \psi \prod_{j=1}^4 (1 - p_j) + (1 - \psi)$$

We can find the Likelihood of all detection probabilities for a species across sites via

$$L(\psi, p | Data) = \prod_{i=1}^4 \Pr(H_i)$$

Observed data likelihood

Assume a detection history of [1,0,0,1]

What is $\Pr(H = 1001)$?

Note: We can accommodate missing data by updating the probability statements to skip missing observation

E.g. [1,0,-,1]

$$\Pr(H = 10 - 1) = \psi * p_1(1 - p_2)p_4$$

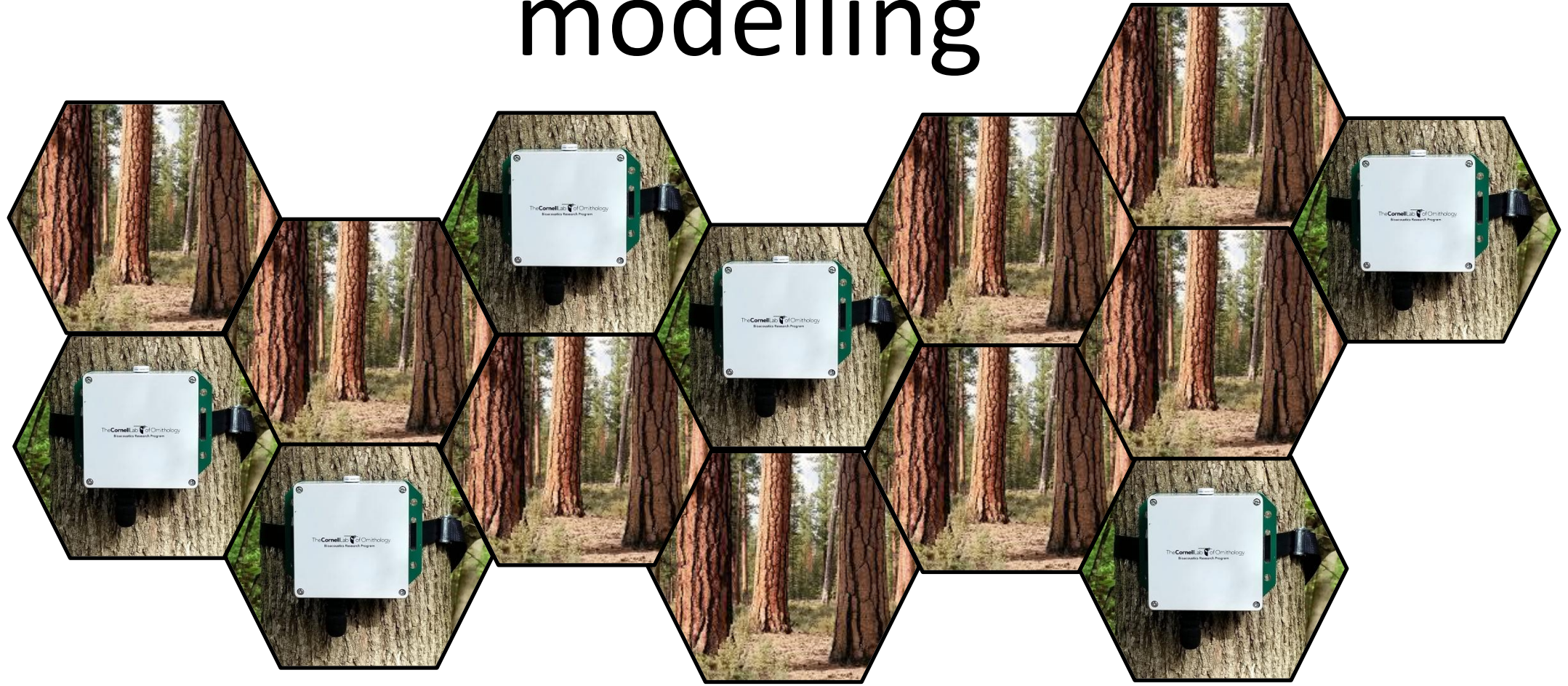
across sites via

$$L(\psi, p | Data) = \prod \Pr(H)$$

Necessary assumptions

1. Occupancy state of a location does not change during survey
2. No mis-identification (no false positives)
3. Detection histories of each location are independent
4. Detection in each survey is independent of other surveys
5. Probability of occupancy is equal across all units
6. Probability of detection is equal across all units
 - Can be relaxed with “*or differences are modeled with covariates*”
 - Testing some can be valuable

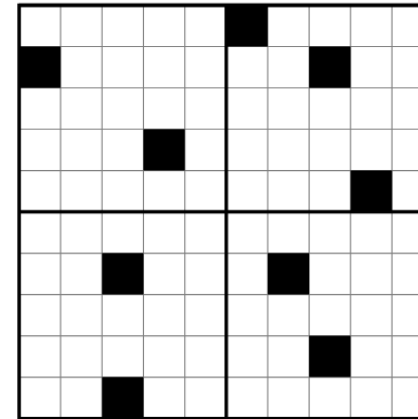
Study design for occupancy modelling



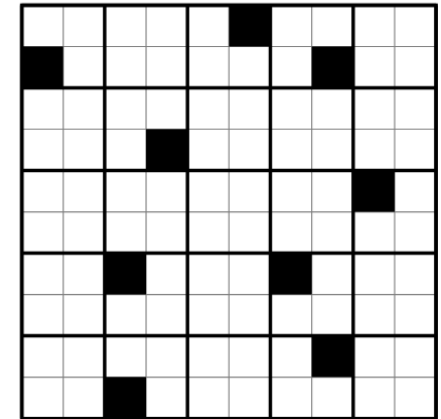
Survey design

- Why? What? **How?**
- Statistical: what type of data can I expect?
- Site selection: where exactly will we be surveying?
- Logistical: can we afford this? Is it safe?
- *Your study/survey design must be consistent with the assumptions of your statistical approach*
 - And your statistical approach must be consistent with the reality of your data

Option A

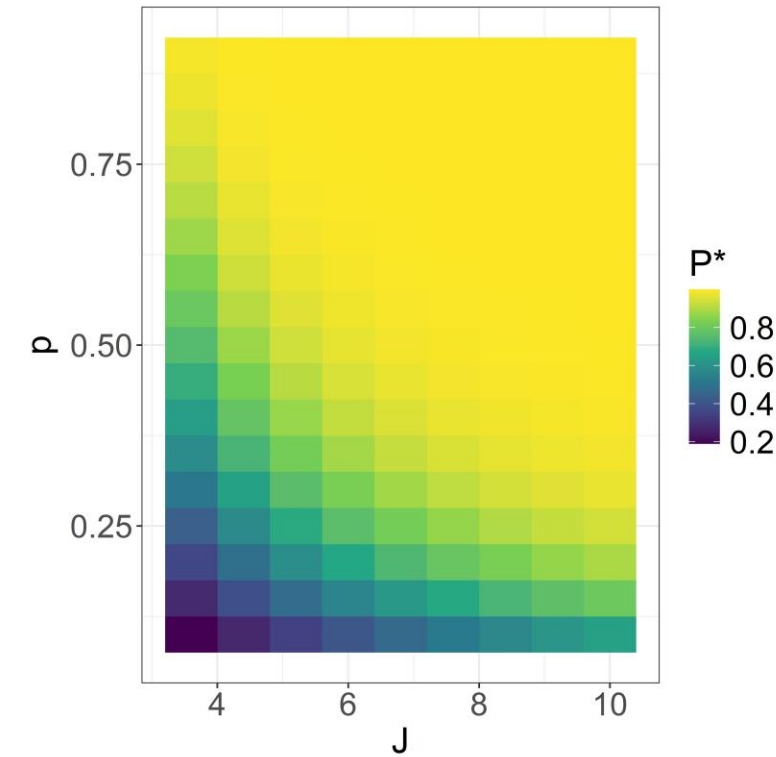


Option B



How long should I sample?

- Entire survey season = primary sampling period (PSP)
- PSP should be:
 1. Doable! (Batteries, SD cards, personnel, etc)
 2. Sufficient to achieve “decent” detection
- Consider starting by doing what you think is too much
 1. Estimate detection
 2. Subsample data, refit models, see how detection changes
 3. (Optional) Simulate trend, test statistical power
- Balance focal species objectives with other species
- Guided by the research objectives



$$P^* = 1 - (1 - p)^J$$

How should I divide my samples into capture histories?

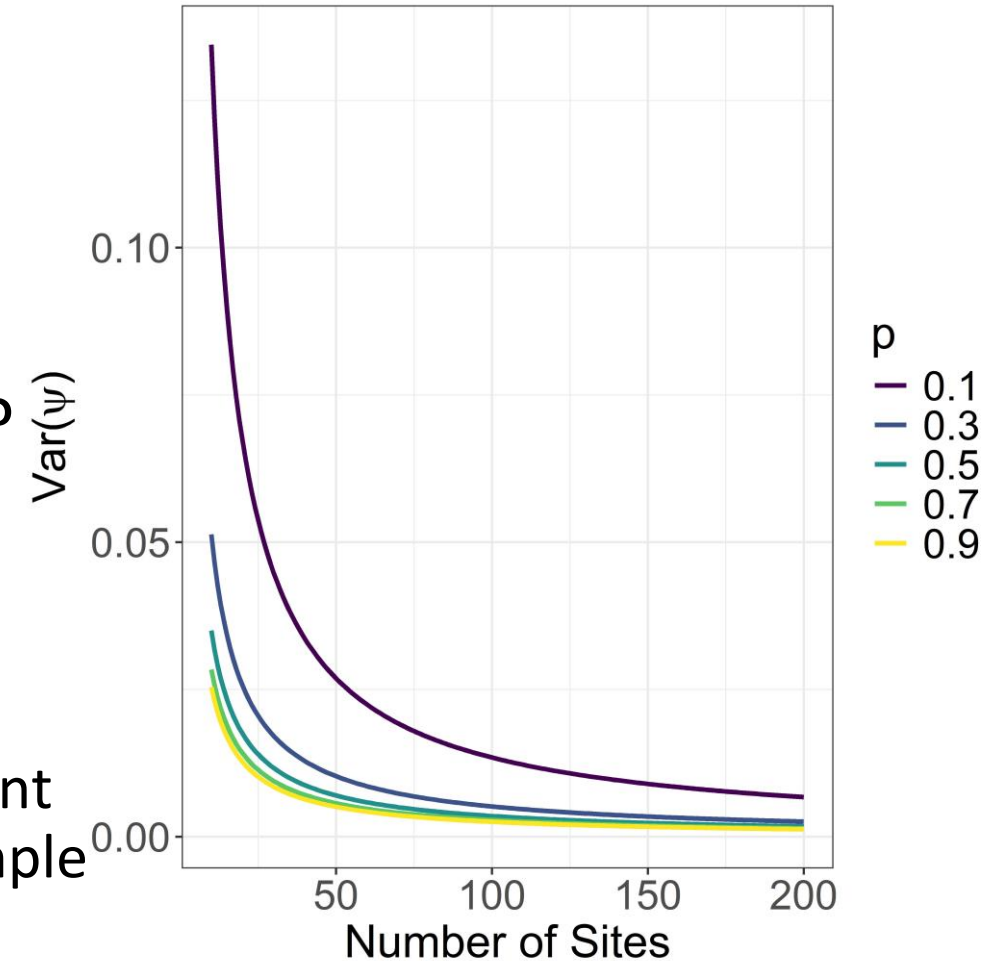
- Consider the ecology of the species
 - How rare is it?
 - How variable do you expect vocalizations to be?
- Consider the limitations of your sampling effort
 - How long can I deploy recorders?
 - When can I access field sites?
- Consider limitations of modelling efforts
 - How much computational power do I have?
 - What style of analysis am I running?
- Consider moving from a “one-size fits all” paradigm
 - de Jager et al., 2026 *Global Ecology and Conservation*

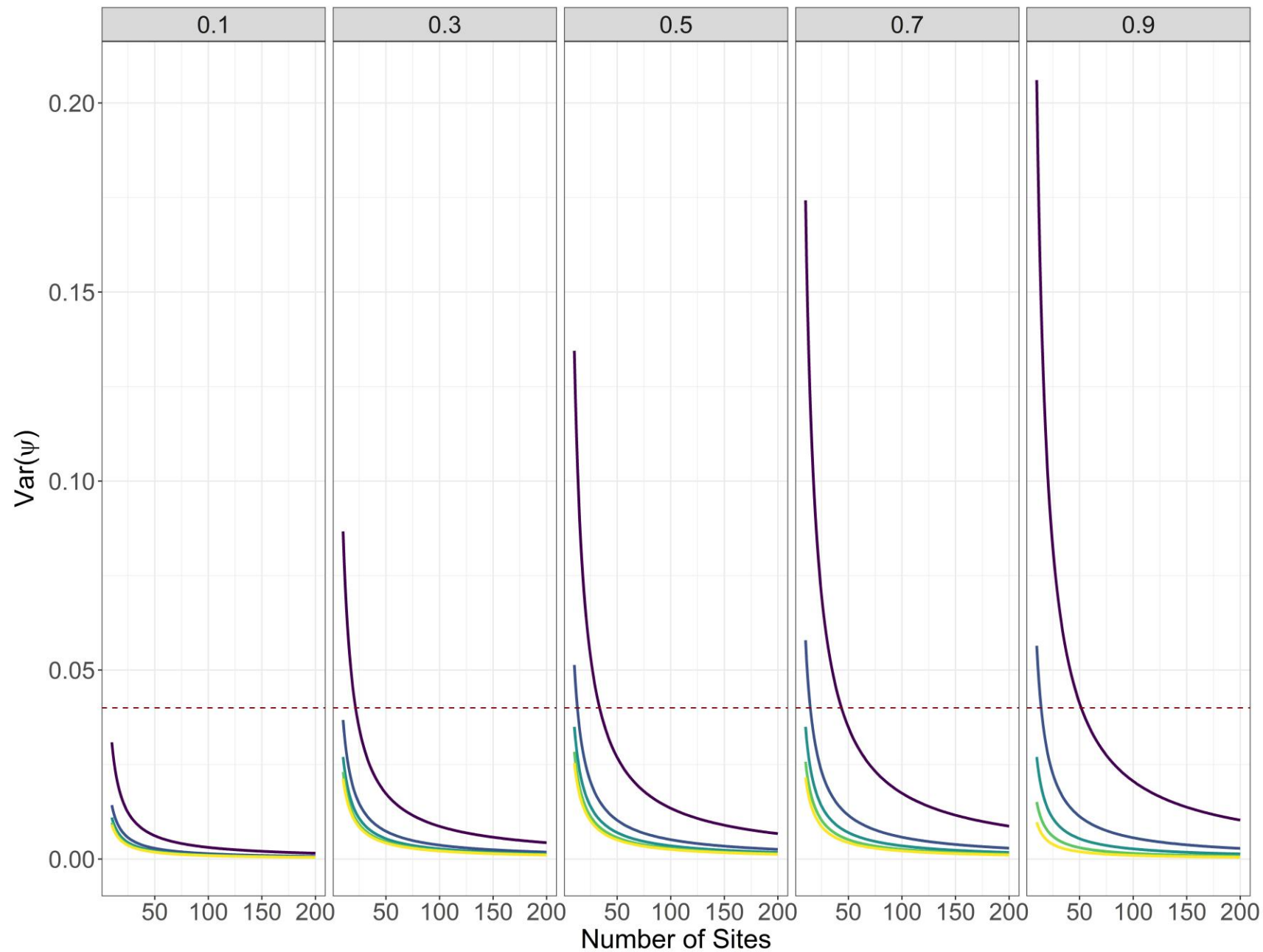
Unit	Design A		
	Day		
	1	2	3
1	x x x		
2			x x x
3		x x x	
4		x x x	
5			x x x
6	x x x		
7		x x x	
8			x x x
9	x x x		
p	0.5	0.3	0.8

Unit	Design B		
	Day		
	1	2	3
1		x	x
2	x x x		x
3	x	x	x
4	x		x
5		x	x
6	x	x	x
7	x		x
8		x	x
9	x	x	x
p	0.5	0.3	0.8

Where should I place my units and how many?

- What are the goals of the study?
 - Are we interested in management...?
 - Species distribution modelling...?
 - Dynamics?
 - Community assessment?
- How large should a sampling unit area be?
 - Large enough to be biologically meaningful
 - Not so large we lose interpretation!
- How far should my units be spaced apart?
 - Enough to ensure independence!
 - Consider the listening range of your equipment and attenuation properties of where you sample



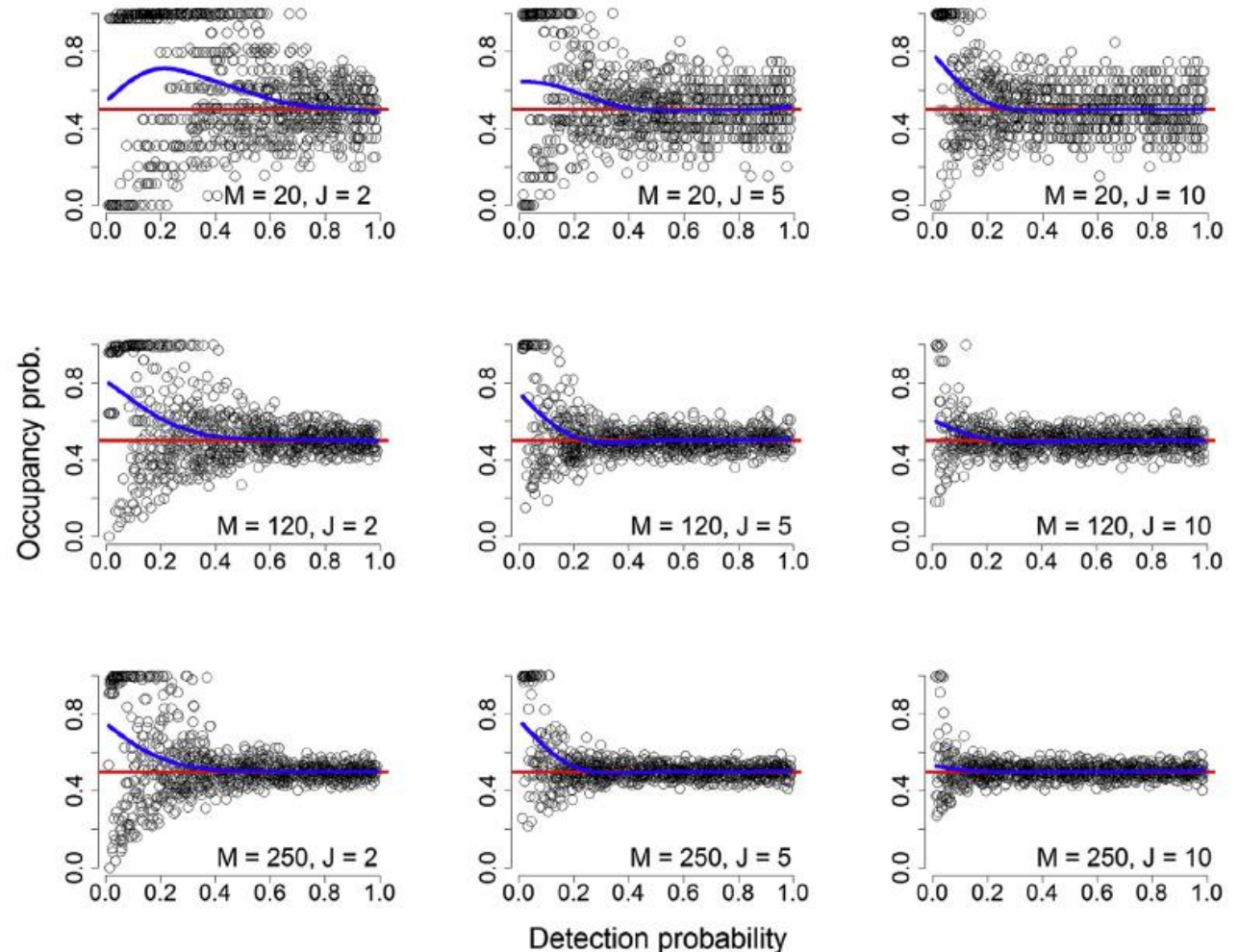


**Higher occupancy →
More sites to achieve
target variance**

**Rare species might
benefit from more
concentrated but longer
sampling**

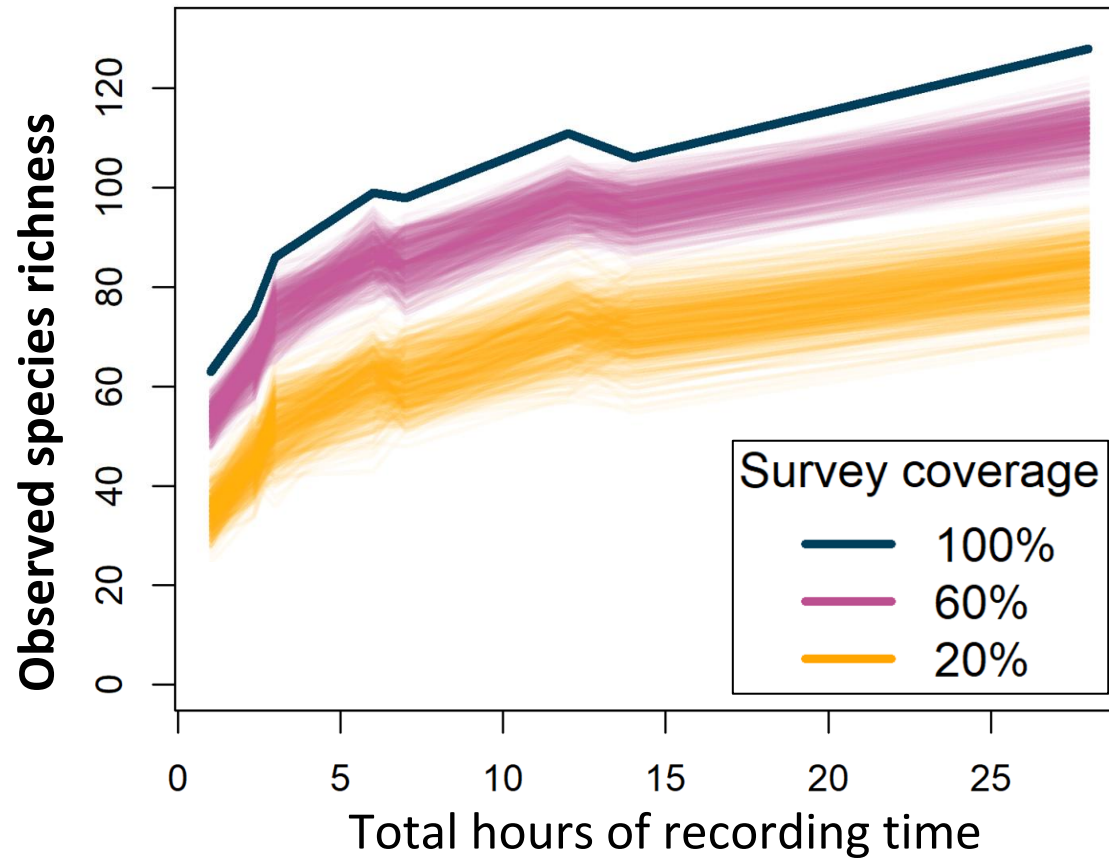
Effort and estimator performance

- Biases at low detection rates
- Greater parameter precision with more replication
- Greater parameter precision with more sites
- Difficult to detect species have higher imprecision (cutoff ~ 0.2)



Rarefaction curves

Northern Sierra Nevada



Wood et al. 2021

CODING BREAK

Let's tackle some simple occupancy models with simulated data

Goals:

- Become familiar with the format of detection histories
- Understand how detection probability and the number of visits and sites influence occupancy estimates

Methods of occupancy modeling for PAM and beyond...

- Frequentist
 - Unmarked
 - R Presence
 - Presence (GUI)
- Bayesian
 - spOccupancy
 - Ubms
 - Custom-based workflow
 - BUGS/JAGS/NIMBLE/STAN

Received: 16 December 2022 | Accepted: 24 April 2023

DOI: 10.1111/2041-210X.14123

APPLICATION

The unmarked R package: Twelve years of advances in occurrence and abundance modelling in ecology

Kenneth F. Kellner¹  | Adam D. Smith²  | J. Andrew Royle³  | Marc Kéry⁴ 
Jerrold L. Belant¹  | Richard B. Chandler⁵ 

Methods in Ecology and Evolution 



Layering on more complexity: Covariates!

- Survey-specific covariates
 - Vary by survey
 - What are some examples?
 - Date, weather, effort, ..., etc!
 - Anything that *can* vary between surveys AND *can* impact detectability!
- Site-specific covariates
 - Vary across space (or time) between sites
 - What are some examples you might use for a species of interest?
 - Habitat composition, configuration, and structure, climate, management intervention, ..., etc!



Layering on more complexity: Covariates!

- Survey-specific covariates
 - E.g., Date and wind
- Site-specific covariates
 - E.g., annual temperature and canopy cover

$$\text{logit}(\psi_i) = \beta_0 + \beta_1 * \text{Temp}_i + \beta_2 * \text{Canopy}_i$$

$$\text{logit}(p_{ij}) = \alpha_0 + \alpha_1 * \text{Date}_{ij} + \alpha_2 * \text{Wind}_{ij}$$

We can also use site-covariates for detection...

$$\text{logit}(p_{ij}) = \alpha_0 + \alpha_1 * \text{Date}_{ij} + \alpha_2 * \text{Wind}_{ij} + \alpha_3 * \text{Canopy Cover}_i$$

Accounting for false-positives in occupancy models

- Derivation of our typical occupancy model
- Finite mixture model
- Royle-Link model estimates parameters w/o validation
- Validation data can improve FP discrimination
 - “Multi-state” occupancy model (Miller et al., 2011)
 - Finite Poisson mixture model (Chambert et al., 2015)

$y_{ij} \sim \text{Bernoulli}(z_i p + (1 - z_i)q)$
Typical Occupancy assumes $q = 0$, so...
 $\Pr(y_{ij} | z_i = 1) \sim \text{Bernoulli}(z_i p)$
but if we expect FPs ($q > 0$)...
 $y_{ij} \sim \text{Bernoulli}(z_i p + (1 - z_i)q)$

Received: 16 July 2021 | Accepted: 18 February 2022

DOI: 10.1111/2041-210X.13905

RESEARCH ARTICLE

Methods in Ecology and Evolution
BRITISH
ECOLOGICAL
SOCIETY

A continuous-score occupancy model that incorporates uncertain machine learning output from autonomous biodiversity surveys

Tessa A. Rhinehart¹  | Daniel Turek²  | Justin Kitzes¹

FP Models: Miller et al., 2011 Multi-state method

Table 7.1 Multi-state observation probabilities for the model of Miller et al. (2011).

True State	Observed State		
	$y = 0$	$y = 1$	$y = 2$
$z = 0$	$1 - p_{10}$	p_{10}	0
$z = 1$	$1 - p_{11}$	$(1 - b) p_{11}$	$b \times p_{11}$

z = Latent State

y = Observed State

p_{10} = FP detection prob; Prob($y = 1|z = 0$)

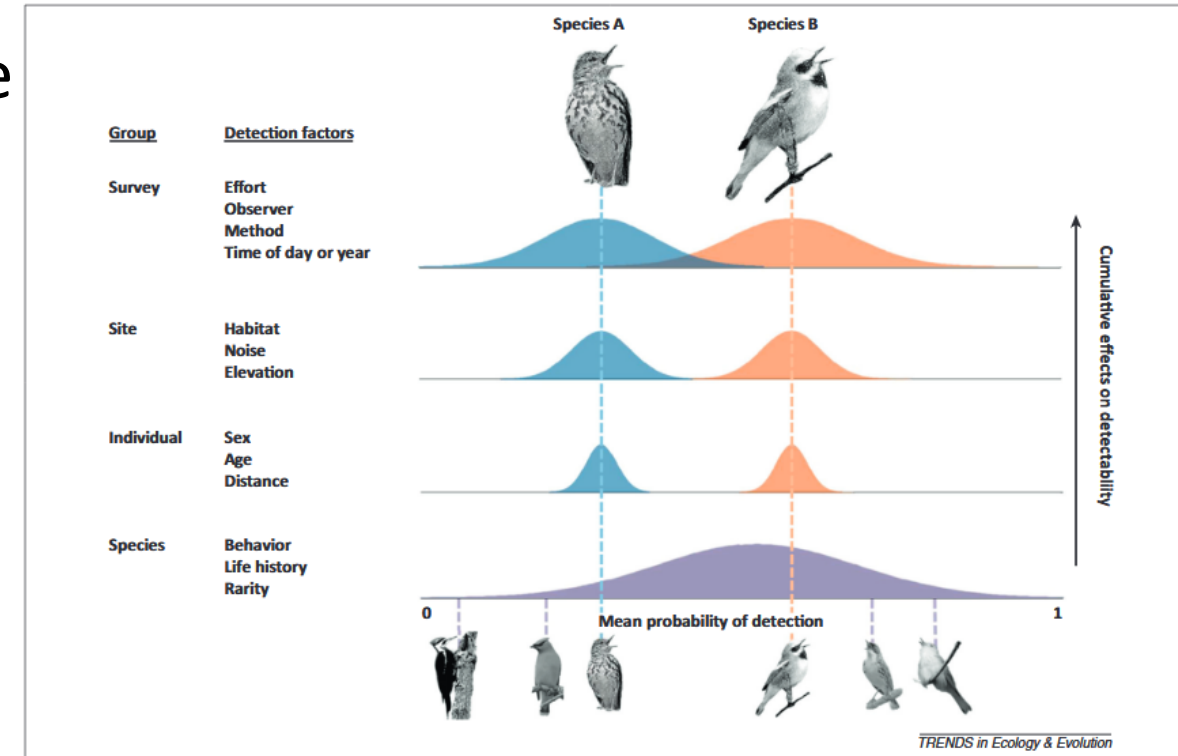
p_{11} = TP detection prob; Prob($y=1|z=1$)

b = Probability a TP detection was designated as “certain”

Sites	Observed	Validated	True
Site A	1	1	1
Site B	0	0	0
Site C	1	0	0
Site D	0	NA	1
Site F	0	NA	0
Site G	1	NA	1

Multi-species, static occupancy model

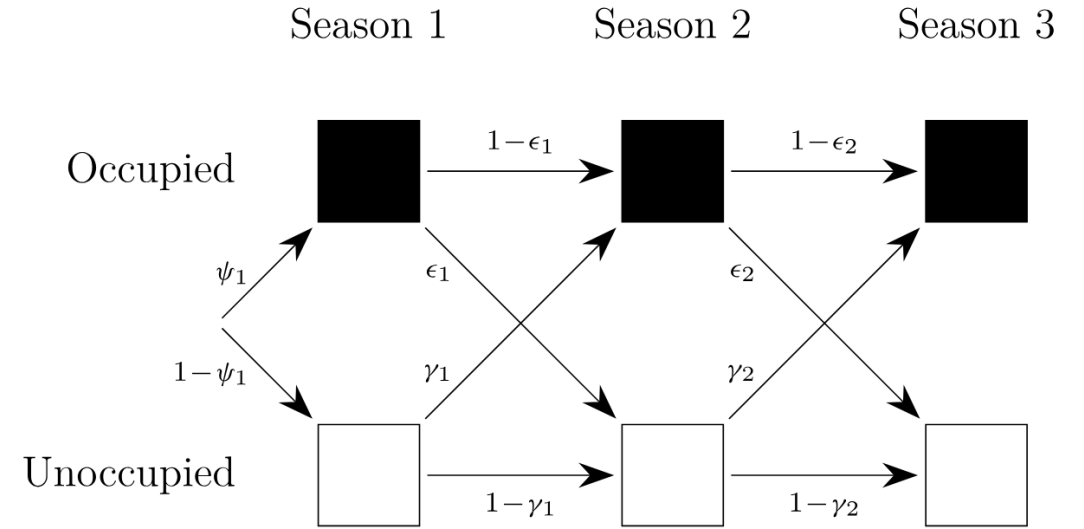
- But, what if we have more than one species of interest?
- Stacked-single species models
- Multi-species occupancy models
- Species detection and occupancy responses come from “community” distribution
 - Efficiency
 - Rare species estimation
 - Meta-community estimates (DA approach; Dorazio and Royle, 2005)



Iknayan et al., 2014

Single-species, *dynamic* occupancy model

- Models transition states over time
 - ϵ = extinction probability
 - γ = colonization probability
- Requires repeat sampling across PSPs
- Data hungry
 - Many sites across many years $\rightarrow$ Colonization/Extinction models
 - Few sites across a few years $\rightarrow$ Autologistic models or stacked season model with random effect



$$z_1 \sim \text{Bernoulli}(\psi_1)$$

$$z_{t+1} | z_t \sim \text{Bernoulli}(z_t(1 - \epsilon) + (1 - z_t)\gamma)$$

$$y_{ij} | z_i \sim \text{Bernoulli}(z_i p)$$

CODING BREAK

Let's get some hands-on experience with fitting an occupancy model to real bioacoustics data

Goals:

- Explore data preparation scripts for formatting data for occupancy models
- Fit single-season, single-species occupancy models with and without covariates
- Interpret coefficient estimate, plot output, multi-model inference, and test model fit
- Explore relaxing assumptions for multi-species, false-positive models, and dynamic models (multi-season)